

Engineering Graduation Project

Cambodia Consumer Price Index (CPI) Pipeline & Inflation Nowcasting

Building an automated data pipeline to scrape, monitor, and nowcast daily inflation in Cambodia.

PRESENTER

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What is the Consumer Price Index (CPI)?

Measuring the Cost of Living

- **What is CPI?** CPI measures the average change over time in the prices paid by consumers for everyday goods and services.
- **The Consumer Basket:** It tracks basic family necessities including rice, meat, vegetables, cooking oil, gasoline, housing, and healthcare.
- **Measuring Inflation:** When the total cost of this basket goes up over time, that rise is called **inflation**.

Why is CPI Important?

- **Guides Economic Policy:** The Ministry of Economy and Finance (MEF) uses CPI to calibrate national budgets, minimum wages, and social safety nets.
- **Protects Currency Stability:** The National Bank of Cambodia (NBC) relies on inflation figures to safeguard the purchasing power of the Khmer Riel.
- **Protects Citizens:** Directly indicates whether life is becoming more expensive for everyday Cambodian families.

The Challenge: Official NIS Publication Delay

⚠️ 30 to 60 Days Publication Delay

- **Manual Data Collection:** Official CPI by the National Institute of Statistics (NIS) relies on survey workers physically visiting retail markets once a month.
- **Substantial Time Lag:** Compiling and validating these manual surveys takes **30 to 60 days** after the end of the month before official numbers are published.
- **Outdated by Arrival:** By the time the official report is released, the economic conditions on the ground have often already changed.

⚠️ Blind to Fast-Moving Price Shifts

- **Vulnerable Cambodian Basket:** Over 70% of typical family spending is in food and fuel—categories where prices fluctuate daily.
- **Intra-Month Blind Spots:** Monthly averages miss sudden price spikes, weekend discounts, and short-term supply disruptions.
- **Delayed Policy Response:** Without real-time data, government interventions and social support arrive too late to protect vulnerable households.

What This Project Aims to Accomplish

1. Build an Automated Data Infrastructure

Design and deploy an end-to-end data pipeline to replace slow manual collection with reliable, automated online extraction and secure storage.

2. Use Airflow to Automate Daily Scraping

Orchestrate 20+ automated web scrapers using Apache Airflow to run every morning at 2:00 AM, capturing over 31,000 daily price quotes reliably.

3. Monitor Scraped Data & Quality

Continuously monitor data freshness, scraper health, missing products, and extreme price anomalies through live operational dashboards.

4. Apply Machine Learning to Nowcast Inflation

Deploy ML models to predict and nowcast current-month inflation in real time, completely bridging the 30-day official NIS publication lag.

How It Works: Simple End-to-End Pipeline

STEP 1

 Scrape

Automated crawlers collect daily prices from 20+ sources: supermarkets (AEON, DeliShop, Chip Mong), pharmacies, fuel stations, bus travel, and official MEF FX rates.

STEP 2

 Bronze Layer

Raw Ingestion: Stores all raw extracted price records, barcodes, and product details exactly as captured, keeping an immutable historical record.

STEP 3

 Silver Layer

Clean & Standardize: Converts USD to Khmer Riel, standardizes units (per kg, per L), matches identical products across stores, and sorts items into official categories.

STEP 4

 Gold Layer

Index & Analytics: Calculates the daily price index using national consumption weights, builds clean summary tables, and feeds the ML nowcasting engine.

What I Am Doing Right Now (Progress & ML)

Active Work

1. Daily Automated Pipeline is Live

Airflow executes scrapers every morning at 2:00 AM, processing >31,000 quotes across Bronze, Silver, and Gold layers.

✓ 100% Live & Running

2. Pipeline & Scraper Health Monitoring

Visual dashboards actively track daily scraper success rates, store coverage, and price anomaly alerts.

✓ 100% Monitored

3. Applying Machine Learning for Inflation Nowcasting

Developing ML models that use fast-moving Food & Fuel signals to forecast the full month's inflation rate ahead of official NIS release.

⚡ Active Model Evaluation

4. Finalizing Thesis Report & Defense Preparation

Benchmarking model accuracy against historical official NIS numbers; preparing for graduation defense on September 30, 2026.

🔍 Final Review